

CV

Amirhosein Shirzad Gavabari

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EDUCATION

Tarbiat Modares University

Tehran, Iran

M.Sc. in Biochemical Engineering

Sep 2022 - Feb 2025

- Supervisors: Dr. Mohsen Nosrati, Dr. Seyed Morteza Zamir
- Thesis: Simultaneous prediction and optimization of methane production and hydrogen sulfide content in an anaerobic digester based on interpretable artificial neural network
- GPA: 18.99/20 (1.38 in German grading system)

University of Tehran

Tehran, Iran

B.Sc. in Chemical Engineering

Sep 2018 - Sep 2022

- GPA: 15.39/20 (2.38 in German grading system)

RESEARCH INTERESTS

- Bioprocess development
- Fermentation
- Bioprocess modeling and optimization
- Enzyme and biopharmaceutical production
- Data analysis and machine learning

PUBLICATIONS

- Shirzad Gavabari, A., Zamir, S. M., Nosrati, M., Interpretable ANN-SHAP framework for multi-objective optimization of CH₄ and H₂S in full-scale anaerobic digestion, Renewable Energy.

LANGUAGE

- Persian: Native proficiency
- English: Full professional proficiency
- German: A1

WORK EXPERIENCE

- 1) Internship, Production Processes Department, Sobhan Daru Co., Jun 2021 - Aug 2021
Tasks & accomplishments: Acquired hands-on insight into pharmaceutical production and, quality control processes, including exposure to manufacturing operations, analytical instruments, and GMP practices.
- 2) R&D Specialist, R&D Department, Bioluence, Feb 2024 - Jan 2025
Tasks & accomplishments: Applied a combination of theoretical research and hands-on laboratory skills to enhance enzyme production in bacteria/yeast strains, involving literature investigation, process data analysis, design of experiments (DoE), and the operation of fermenters and tangential flow filtration (TFF) equipment.
- 3) Up-stream Engineer, Production Department, Arena Biogene, Mar 2025 - Present
Tasks & accomplishments: Conducted upstream bioprocessing for mammalian cell culture from 125 mL shake flasks to 200 L single-use wave bioreactors to produce biopharmaceutical, performing aseptic cell culture, media preparation, cell counting, viability assessment, process monitoring, and supporting scale-up activities under GMP-oriented practices in clean room.

TEACHING EXPERIENCE

- Teaching assistant, Industrial microbiology, Tarbiat Modares University, Fall 2023
Course professor: Dr. Seyed Abbas Shojaosadati

PROJECTS

- 1) Exploring new methods for natural gas sweetening, University of Tehran, May 2022 – Jun 2022
- 2) Feasibility study and investigation of energy recovery from sludge, Mashhad WWTP, Fall 2023
- 3) Design and analysis of downstream process of oxalic acid bio-production using SuperPro Designer, University of Tarbiat Modares, In progress

SKILLS

Technical Skills

- **Programming & data Science:** Proficient in Python for machine learning, deep learning, and data analysis.
- **Process Optimization:** Experienced in applying metaheuristic algorithms and statistical approaches for process improvement.
- **Modeling & simulation:** Skilled in process modeling and simulation using SuperPro Designer.
- **Statistical Analysis:** Proficient in design of experiments (DoE) for process statistical characterization and optimization.

Software Proficiency

- **Specialized Software:** SuperPro Designer, Design Expert, Qualitek-4
- **Development Tools:** Google Colab
- **Office Suite:** MS Office (Advanced Excel, Word, and PowerPoint)

Laboratory Skills

- **Bioprocessing Equipment:** Experience in fermenter and single use bioreactor operation and monitoring, depth filters and, tangential flow filtration (TFF).
- **Analytical Techniques:** Hands-on experience in aseptic cell culture, media preparation, cell counting, viability assessment, using UV-vis spectroscopy and high-performance liquid chromatography (HPLC). Also familiar with SDS-PAGE, PCR and gel-electrophoresis
- **GxP:** familiar with GMP and GLP guidelines. Experience in working at cleanrooms

Soft skills:

- Teaching, communication, teamwork, problem solving, innovation, adaptability.

LICENSES AND CERTIFICATES

- 1) Algae Biotechnology, Coursera, Aug 2023
- 2) Introduction to Industrial Bioprocess Development, Coursera, Jul 2023
- 3) Industrial Biotechnology, Coursera, Jun 2023
- 4) HSE basics, Iranian Association of Chemical Engineering (IACHE), Dec 2020
- 5) Introduction of Matlab, University of Tehran, Aug 2019
- 6) MS Excel for data analysis, University of Tehran, Aug 2019

HONORS AND AWARDS

- 1) Ranked within the top 5% among approximately 144,000 participants in the National Entrance Examination for Iranian Universities, Iran
- 2) Ranked as the 1st student among all students of biochemical engineering at the end of the master's, Tarbiat Modares University, Iran
- 3) Awarded a competitive research grant by the Iran National Science Foundation (INSF), Project No. 4031210

LEADERSHIP POSITIONS

- Secretary of the Tarbiat Modares biochemical and biomedical engineering scientific association, Nov 2023 - Nov 2024

REFERENCES

Dr. Mohsen Nosrati
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Tarbiat Modares University
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